

# MD TANVIR ISLAM

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## EDUCATION AND AWARDS

|                              |                                                                                                                                                                                                                                                    |                            |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| February 2023<br>August 2025 | <b>M.Sc., SUNGKYUNKWAN UNIVERSITY (SKKU), South Korea</b><br><i>Department of Computer Science and Engineering   CGPA : 4.19/4.50</i><br>Thesis : Study Towards Robust Image Enhancement in Adverse Atmospheric Environments.                      | <a href="#">Transcript</a> |
| March 2022<br>February 2023  | <b>Korean Language and Cultural Program, PAI CHAI UNIVERSITY, South Korea</b><br><i>Korean Language and Cultural Program   TOPIK Level : 4/6</i><br>As a prerequisite, I completed this program under the Global Korea Scholarship (GKS).          | <a href="#">Transcript</a> |
| January 2015<br>January 2019 | <b>B.Sc., North Western University (NWU), Bangladesh</b><br><i>Department of Computer Science and Engineering   CGPA : 3.98/4.00</i><br>Thesis : Study on diabetes prediction for typical and non-typical cases using machine learning approaches. | <a href="#">Transcript</a> |

## RESEARCH EXPERIENCES

IF : IMPACT FACTOR

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| March 2023<br>Present    | <b>Research Assistant, VIS2KNOW LAB, Sungkyunkwan University, South Korea</b><br><a href="#">Lab Director</a> <ul style="list-style-type: none"><li>➤ Data curation and research in AI image detection, image dehazing, and low-light image enhancement.</li><li>➤ Our research on image dehazing, LLIE, and AI-synthesized image detection has led to three papers are published at top-tier conferences and journals such as <i>ACM Multimedia (Rank A)</i>, <i>Asian Conference on Computer Vision (ACCV) (Rank B)</i>, <i>Alexandria Engineering Journal (AEJ) (IF : 6.2, Q1)</i>. Also, multiple research papers and patents are under review by the <i>CVPR'24</i> and the <i>US Patent Office</i>.</li></ul> |
| July 2018<br>August 2021 | <b>Research Assistant, CSHI-RLAB, North Western University, Bangladesh</b><br><a href="#">Lab Director</a> <ul style="list-style-type: none"><li>➤ Collected real-life datasets to perform research using machine-learning techniques and <b>published several papers</b> at different international conferences, which achieved around <b>100 citations</b> so far.</li></ul>                                                                                                                                                                                                                                                                                                                                      |

## TEACHING EXPERIENCES

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| March 2024<br>August 2024     | <b>Teaching Assistant, DEPARTMENT OF AI, Sungkyunkwan University, South Korea</b> <ul style="list-style-type: none"><li>➤ Assisted with curriculum planning for courses and held office hours to support and mentor students.</li><li>➤ Facilitated group discussions and study sessions to enhance collaborative learning.</li></ul> |
| January 2019<br>December 2020 | <b>Part-time Lecturer, SOUTH-EAST ENGINEERING COLLEGE, Khulna, Bangladesh</b> <ul style="list-style-type: none"><li>➤ Planned, prepared and presented lectures while assessing student assignments.</li><li>➤ Maintained communication with teachers and students.</li></ul>                                                          |

## AWARDS AND SCHOLARSHIPS

|                              |                                                                                                                                                                                                       |                             |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| November 2024                | <b>Academic Excellence Award Winner, NIIED, Government of South Korea</b><br>The NIIED selected me as one of the three students for this award based on the research outcomes.                        | <a href="#">Certificate</a> |
| December 2023                | <b>16th Korean Language Society Scholarship, PAI CHAI UNIVERSITY, South Korea</b><br>Received this award based on my academic performance and on-campus activities including volunteering.            | <a href="#">Certificate</a> |
| August 2022<br>February 2025 | <b>Global Korea Scholarship (GKS), NIID, South Korea</b><br>Received this award based on my academic performance and on-campus activities including volunteering.                                     | <a href="#">Certificate</a> |
| January 2015<br>January 2019 | <b>First Position in Merit List, NORTH WESTERN UNIVERSITY, Bangladesh</b><br>Certified by my undergraduate university for securing the 1 <sup>st</sup> position in the merit list among all students. | <a href="#">Certificate</a> |

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|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCIE   Q1   IF : 6.2 | <b>Md Tanvir Islam</b> , Ik Hyun Lee, Ahmed Ibrahim Alzahrani, and Khan Muhammad*. "MEXFIC : A meta ensemble eXplainable approach for AI-synthesized fake image classification." <i>Alexandria Engineering Journal</i> 116 (2025) : 351-363. (Published)                                             |
| SCIE   Q1   IF : 7.8 | <b>Md Tanvir Islam</b> , Safkat Shahrier Swapnil, Md Masum Billal, Asif Karim*, Niusha Shafiabady, and Md Mehedi Hassan. "Resource constraint crop damage classification using depth channel shuffling." <i>Engineering Applications of Artificial Intelligence</i> 144 (2025) : 110117. (Published) |
| Rank A               | <b>Md Tanvir Islam</b> , Nasir Rahim, Saeed Anwar, Muhammad Saqib, Sambit Bakshi, and Khan Muhammad*. "Hazespace2m : A dataset for haze aware single image dehazing." In Proceedings of the <i>32nd ACM International Conference on Multimedia</i> , pp. 9155-9164. 2024. (Published)                |
| Rank B               | <b>Md Tanvir Islam</b> , Inzamamul Alam, Simon S. Woo*, Saeed Anwar, I. K. Lee, and Khan Muhammad*. "LoLI-Street : Benchmarking Low-Light Image Enhancement and Beyond." In Proceedings of the <i>Asian Conference on Computer Vision</i> , pp. 1250-1267. 2024. (Published)                         |
| Rank A               | Inzamamul Alam, <b>Md Tanvir Islam</b> , and Simon S. Woo*. "Saliency-Aware Diffusion Reconstruction for Effective Invisible Watermark Removal." <i>ACM Web Conference, WWW 2025</i> . (Accepted)                                                                                                    |
| Rank A               | Inzamamul Alam, <b>Md Tanvir Islam</b> , Simon S. Woo*, Khan Muhammad. "WaveGuard : Advancing Invisible Watermarking with Spectral Projection." <i>Computer Vision and Pattern Recognition, CVPR 2025</i> . (In Review)                                                                              |
| U.S. Patent          | <b>Md Tanvir Islam</b> , Ik Hyun Lee, Abdul Khader Jilani Saudagar, Abdullah Altameem, Mohammed Abaoud, and Khan Muhammad. "Apparatus and method for single image dehazing considering haze type classification." <i>U.S. Patent No. 12,217,398</i> , issued February 4, 2025. (Registered)          |
| Book Chapter         | Rezuana Haque and <b>Md Tanvir Islam</b> (2024). "Neural Dynamics : Unravelling the Complexity of Brain Activity." In <i>Brain Networks in Neuroscience : Personalization Unveiled Via Artificial Intelligence</i> . Chapter 1, pp. 1-19. (In review)                                                |

## SKILLS AND INTERESTS

|             |                                                                                             |
|-------------|---------------------------------------------------------------------------------------------|
| Language    | Bengali (Native), English (IELTS (2019) : 7.0, no band < 6.5), Korean (TOPIK (2022) : 4/6). |
| Programming | Python; Toolkit : PyTorch, Keras; Database : PostgreSQL, MySQL.                             |
| Tools       | PowerBI, LaTeX, PyCharm, Jupyter, VSCode, Photoshop, Illustrator, Figma, Microsoft Suites.  |
| Interests   | Image Restoration, Invisible Watermarking, Activity Recognition, Deepfake, Bioinformatics.  |

## TRAININGS AND LEARNING

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                  |
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| August 2019<br>September 2019 | LICT Top-up IT Training, <b>ICD DIVISION</b> , Bangladesh<br><i>Bangladesh Computer Council (Certified by George Washington University, Washington, D.C.)</i><br><ul style="list-style-type: none"> <li>▶ The training includes 240 hours of technical skills such as Java, HTML, CSS, JavaScript, Bootstrap, and MVC Pattern, and 60 hours of other soft skills like presentation and communication.</li> </ul>                                                                                                                                    | Certificate                                                                                                                                                                      |
| February 2018                 | Google Bangladesh Android StudyJam, <b>GDG DHAKA</b> , Bangladesh<br><i>ICT Division, GDG Dhaka, Google Developers</i><br><ul style="list-style-type: none"> <li>▶ A 225-hour Android app development course with regular mini-project assignments, culminating in a final complete app development project.</li> </ul>                                                                                                                                                                                                                             | Certificate                                                                                                                                                                      |
| January 2019<br>Present       | Self Learning,<br><i>Some of the courses and books I completed to grow my knowledge.</i><br><ul style="list-style-type: none"> <li>▶ PyTorch for Deep Learning Bootcamp (52 hours)</li> <li>▶ Python for Data Science, AI &amp; Development (25 hours)</li> <li>▶ Python for Data Science and Machine Learning Bootcamp (25 hours)</li> <li>▶ Foundation of User Experience (UX) Design (19 hours)</li> <li>▶ Modern Computer Vision with PyTorch</li> <li>▶ Python Deep Learning Projects</li> <li>▶ Advanced Deep Learning with Python</li> </ul> | Online, Books.<br>Udemy   Mar 2024<br>IBM   Mar 2024<br>Udemy   Sept 2019<br>Google   Oct 2021<br>Book by V Kishore Ayyadevara<br>Book by Matthew Lamons<br>Book by Ivan Vasilev |

## PROJECTS

### HAZESPACE2M – [GITHUB.COM/TANVIRNWU/HAZESPACE2M](https://github.com/TANVIRNWU/HAZESPACE2M)

1<sup>st</sup> LEAD AUTHOR

A large-scale hazy dataset of over 2 Million images across 10 distinct levels and propose a novel framework for haze-aware single image dehazing that significantly improves PSNR by 2.41%, SSIM by 17.14%, and MSE by 10.2% over general dehazers.

[SOTA Image Dehazing Methods](#) [SOTA Classifiers](#) [PyTorch](#) [Overleaf](#) [PyCharm](#) [Photoshop](#)

### TriFUSE & LOLI-STREET – [GITHUB.COM/TANVIRNWU/TriFUSE](https://github.com/TANVIRNWU/TriFUSE)

1<sup>st</sup> LEAD AUTHOR

First to introduce a 33K paired low-light street scene dataset with object detection labels and propose TriFUSE, a transformer-wavelet-diffusion model that achieves SOTA results for real-life low-light image enhancement for autonomous systems.

[LLIE Methods](#) [Diffusion Denoising](#) [Transformers](#) [Deep Learning](#) [PyTorch](#) [Overleaf](#) [PyCharm](#) [Photoshop](#)

### SADRE – [GITHUB.COM/TANVIRNWU/SADRE](https://github.com/TANVIRNWU/SADRE)

2<sup>nd</sup> LEAD AUTHOR

We introduced SADRE which is a saliency-aware diffusion reconstruction method for effective invisible watermark removal.

[SOTA Image Watermarking Methods](#) [Attacks](#) [Diffusion](#) [PyTorch](#) [Overleaf](#) [PyCharm](#) [Illustrator](#)

## OTHER ACTIVITIES

|                                                                                                                     |                     |
|---------------------------------------------------------------------------------------------------------------------|---------------------|
| Conference Paper Reviewer, <a href="#">International Conference of Pattern Recognition (ICPR)</a> , Kolkata, India. | July 2024           |
| Journal Paper Reviewer, <a href="#">PLOS One</a>                                                                    | January 2025        |
| Journal Paper Reviewer, <a href="#">PLOS Digital Health</a>                                                         | November 2024       |
| Journal Paper Reviewer, <a href="#">Frontiers in Nutrition</a>                                                      | March 2024          |
| Founder of Animal Rescue Team, <a href="#">Khulna Cat Society</a> , Khulna, Bangladesh                              | July 2016 - Present |
| Paper Presentation at <a href="#">ICCISIoT Conference</a> , NIT Agartala, India                                     | Dec 2019            |

## REFERENCES

Khan Muhammad | Associate Professor

Dept. of Applied AI, SUNGKYUNKWAN UNIVERSITY, SOUTH KOREA

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